

## Report Analysis: Image Enhancement

### A. Blur

Original



Sharpened Filter



Histogram Equalization



Original image was a self picture whilst moving causing the motion blur as such a blur image. Notable features to take note is the face, glasses, blanket and the curtain behind the person in image. Initial step after greyscaling is to sharpen the image first using the embossment kernel as:

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

After the embossment filter then applying Histogram Equalization. At initial sharpen filter (embossment) has a noticable increase in definition of curtain patterns as well as more distinct blanket texture. Next, histogram equalization is applied that spreads out the pixel brightness values. The resulting image is a more defined image that decreases the blur of image and able to define more of the textures in the image

## B. High Brightness

Original



Gamma Correction



Original



Gamma Correction



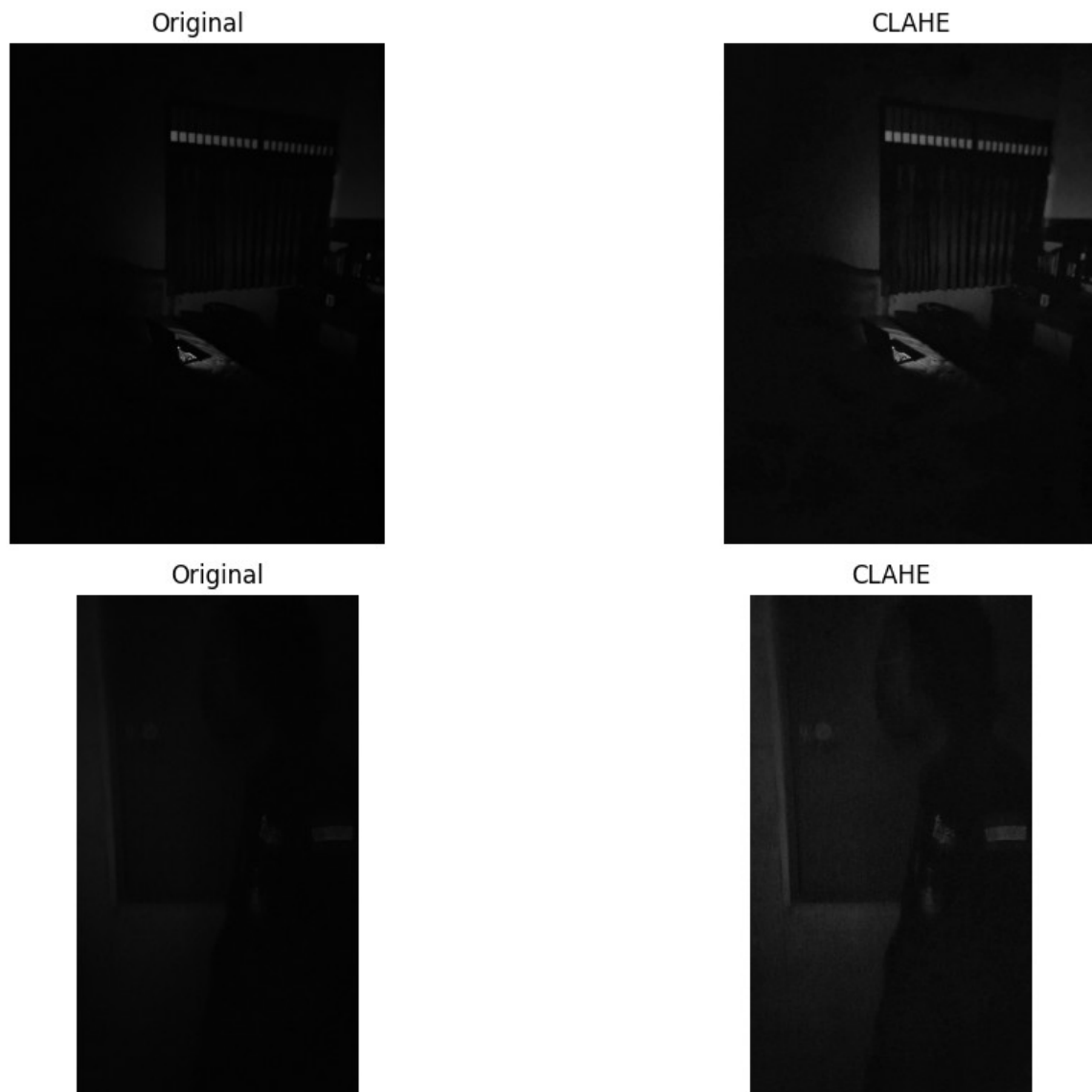
The image used is a market that is relatively bright even after greyscaling. Most notably the contents of the square in the upper middle part of image is barely readable (the one next to the neon writing of wan). The method use to address the high brightness is via gamma correction.

Gamma correction (Power-Law Transformation) is a non linear operation that adjusts pixel brightness based on the formula  $s = c * r^{\gamma}$ . Depending on what number the gamma is set to will effect the output of the correction. If gamma = 1, no change will take effect. If gamma < 1, it brigthens dark shadow regions. If gamma > 1, it compresses bright areas into darker levels.

The usage of gamma = 4.0 is used as original image contains a noticable overexposed bright areas and the drastic number of gamma allows the contents to be more visible by darkening the brighter areas. It is now possible to discern the words in the box the words Super Hemat where previously was difficult to read compared to after gamma correction. Other notable effects from the gamma correction is such as more noticable floor tile details and neon light shape outlines, showing effectiveness for bright/overexposure parts while sufficient light will be darker and loss details

A 2<sup>nd</sup> image at a bright square when applying the gamma correction shows a different view as in a generally brighter image will result in much more areas to be darken to compensate the correction. This overall shows that gamma correction to a more noticable degree could be effective to not effective, relative to how evenly bright overall base image is pre correction.

### C. Dark Image



For the dark image enhancement take 2 cases, 1<sup>st</sup> image is a dimly lit room where light source is only from light of laptop. Another image being a barely lit room with a person inside it. The method used is Contrast Limited Adaptive Histogram Equalization (CLAHE) over regular histogram equalization. This approach is used to avoid the over aggressive stretching of regular histogram equalization.

Applying the CLAHE filter improves the quality of the image. For image 1 of the dimly lit room, there is a slight brightness and more defined image in able to see the curtains and a more noticable shadow of the desk. In the 2<sup>nd</sup> image where it has no noticable lightsource and such barely no noticable details in original image. After applying CLAHE, the image is clarified and there is a shadow outline of a person, showing the effectiveness of the program. CLAHE works by dividing image into grids of small regions (where in the base code, the tile size is 8,8) then performs histogram equalization of each tiles. In addition to the CLAHE, a contrast treshold was set to clip any histogram equalization that exceeds the limit and redistribute evenly before the equalization takes place.

#### D. Low Contrast

Original Low Contrast



Stretching (2%-98% Clip)



Original base image shows passenger side of a car, containing a bright seat cushion, a leather bag on the floor, and a floor mat with a pattern. The image is low contrast as all the details of image share similar tones as the dark floor colour and the bag looks blended together.

The method used to enhance the image is contrast stretching but using min-max contrast stretching has a possibility to fail as some areas are already naturally pure black (0). To address this, a percentile based contrast stretching was applied at minimum 5% and maximum 95%. This means that it ignores the 5% darkest and brightest pixels, to focus only on the main image data. Spreading the full values (0-255) helps define fold of bag, seat cushion more visible and texture of floor mat more defined without creating any odd visual effects from the enhancement